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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

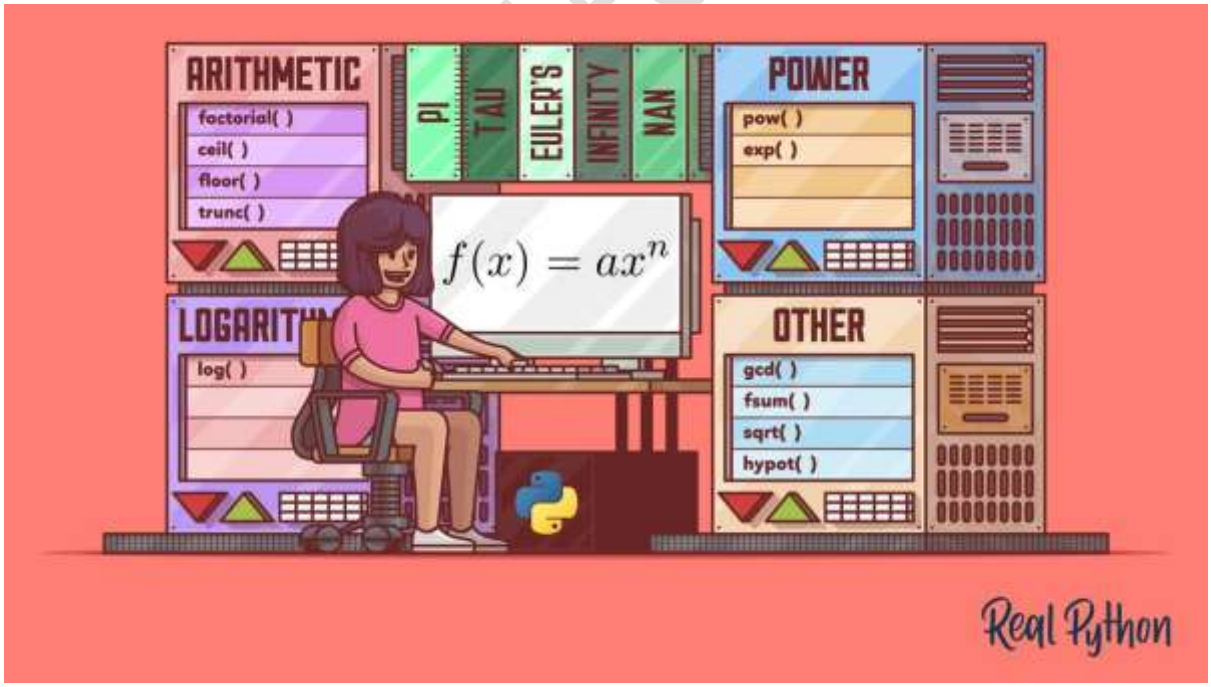
Classes : Offline & Online

Daily Teaching : 2 Hours

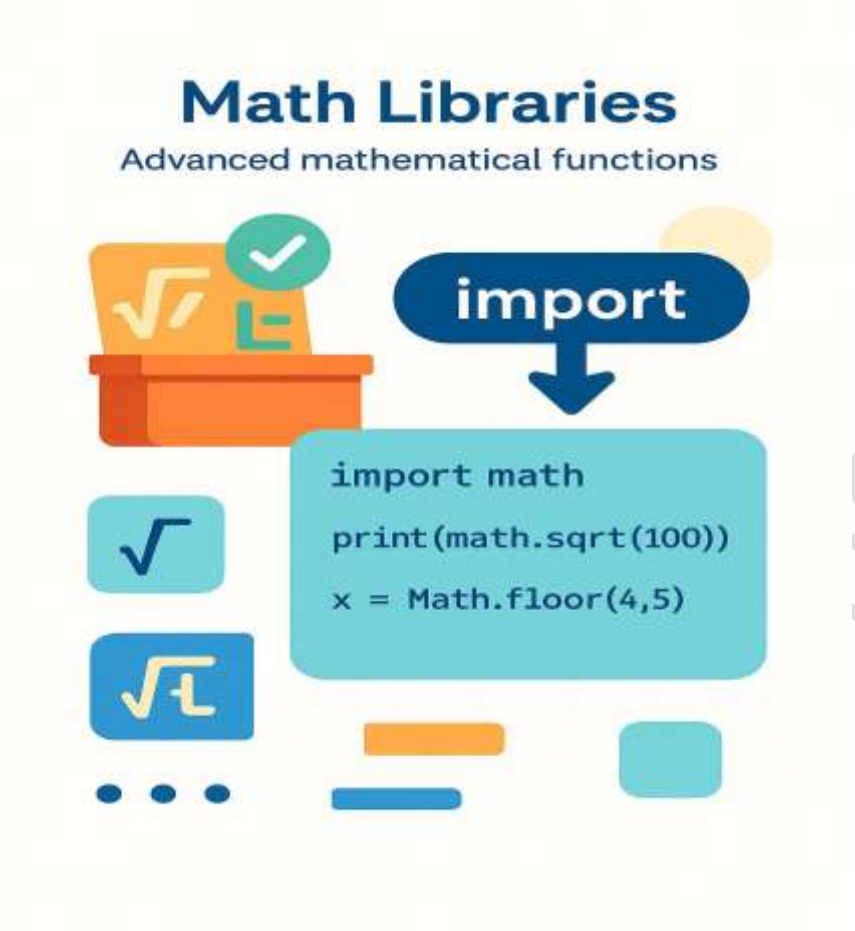
Total Classes : 100

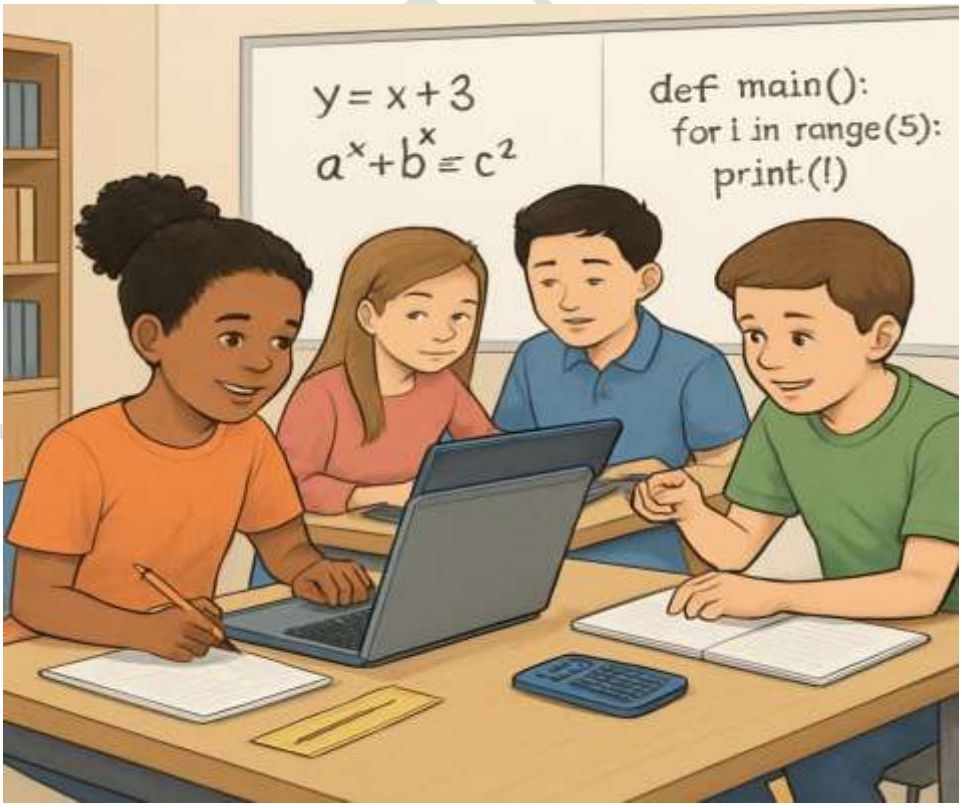
Level 2 – Young Technologist (7th–8th Standard)

Class 35: Math Module in Python



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Math Libraries
Advanced mathematical functions
import
import math
print(math.sqrt(100))
x = Math.floor(4,5)


y = x + 3
a^x + b^x = c^2
def main():
for i in range(5):
print(!)

Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a **Python module** is.
- Understand the **Math Module**.
- Import the math module.
- Find square roots.
- Find powers.
- Round numbers up and down.
- Work with mathematical constants.
- Use mathematical functions in simple programs.
- Create small math-based Python projects.

1. What is a Module?

A **module** is a collection of ready-made Python code that provides useful functions.

Think of a module as a **toolbox** .

Instead of writing everything from the beginning, we can use tools that Python already provides.

Examples of Python Modules

Module	Used For
random	Random numbers and choices
math	Mathematical calculations
turtle	Drawing and graphics
datetime	Date and time

Today we will learn the **Math Module**.

2. What is the Math Module?

Python's **math module** provides many useful mathematical functions and constants.

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It can help us perform calculations such as:

- √ Square root
- Powers
- Rounding
- Factorials
- Greatest Common Divisor
- Trigonometry
- Mathematical constants

To use it, we write:

import math

3. Importing the Math Module

import math

Meaning

import → Bring a module into our program.

math → Name of the mathematical module.

So:

import math

means:

"Python, give me the mathematical tools available in the math module."

4. math.sqrt()

sqrt() means **square root**.

Example

import math

answer = math.sqrt(25)

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print(answer)

Output

5.0

Because:

5 × 5 = 25

More Examples

import math

print(math.sqrt(9))

print(math.sqrt(16))

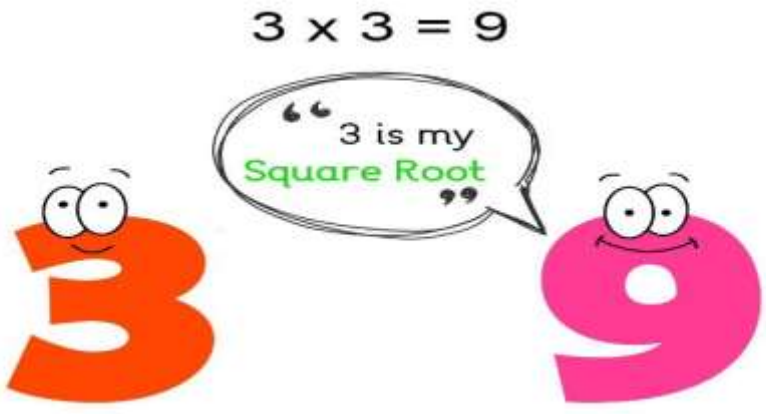
print(math.sqrt(100))

Output:

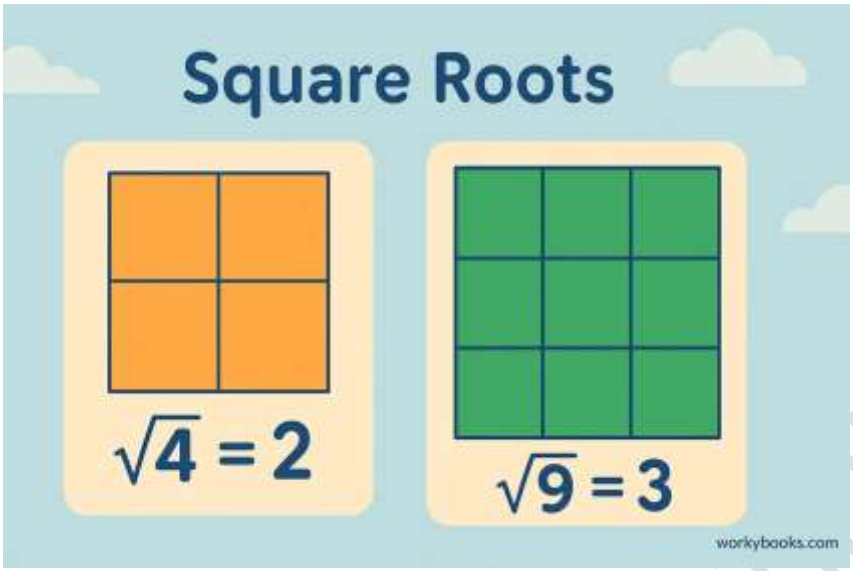
3.0

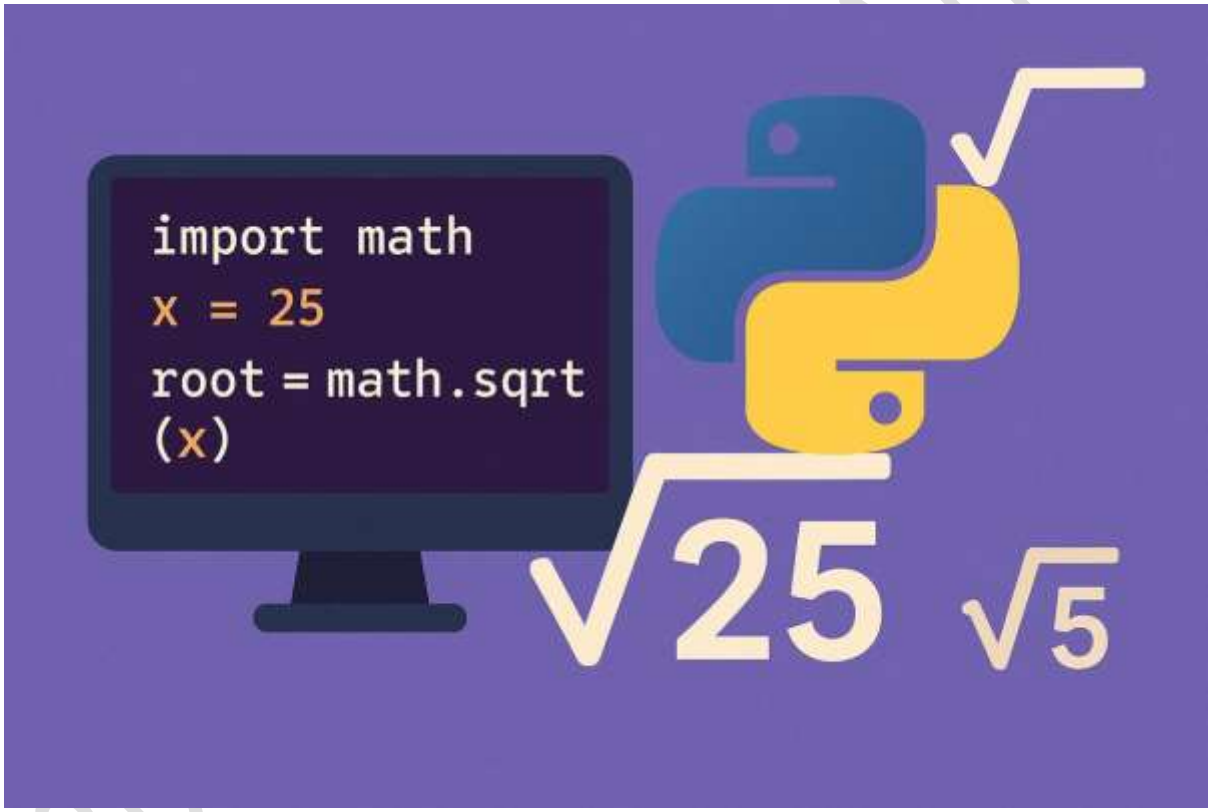
4.0

10.0



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5. math.pow()

pow() is used to calculate powers.

Example

import math

```
answer = math.pow(2, 3)


print(answer)
```

Output:

8.0

Because:

2 × 2 × 2 = 8

More Examples

```
print(math.pow(5, 2))

print(math.pow(3, 3))

print(math.pow(10, 2))
```

Output:

25.0

27.0

100.0

6. math.ceil()

ceil() rounds a number **up** to the nearest whole number.

Example

```
import math


print(math.ceil(4.2))
```

Output:

5

Another example:

```
print(math.ceil(7.1))
```

Output:

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8

Remember

ceil = ceiling = go UP ⬆️

7. math.floor()

floor() rounds a number **down** to the nearest whole number.

```
import math


print(math.floor(4.9))
```

Output:

4

Another example:

```
print(math.floor(8.7))
```

Output:

8

Remember

floor = ground = go DOWN ⬇️

Easy Way to Remember

ceil() → ⬆️ UP

floor() → ⬇️ DOWN

Number	ceil()	floor()
4.2	5	4
5.8	6	5
7.1	8	7
9.9	10	9

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8. math.factorial()

A factorial is written using !.

For example:

5! = 5 × 4 × 3 × 2 × 1 = 120

Python can calculate this using factorial().

```
import math


print(math.factorial(5))
```

Output:

120

More Examples

```
print(math.factorial(3))

print(math.factorial(4))

print(math.factorial(6))
```

Output:

6

24

720

9. math.gcd()

gcd() means **Greatest Common Divisor**.

It finds the largest number that can divide two numbers without a remainder.

Example:

```
import math


print(math.gcd(12, 18))
```

Output:

6

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Because:

- Factors of 12 → 1, 2, 3, 4, 6, 12
- Factors of 18 → 1, 2, 3, 6, 9, 18

The greatest common factor is **6**.

10. math.lcm()

lcm() means **Least Common Multiple**.

It finds the smallest positive number that is a multiple of both numbers.

import math

print(math.lcm(4, 6))

Output:

12

Because:

- Multiples of 4 → 4, 8, **12**, 16...
- Multiples of 6 → 6, **12**, 18...

So the LCM is **12**.

11. math.pi

math.pi gives the mathematical value of **π (pi)**.

The approximate value is:

3.14159

Example:

import math

print(math.pi)

Possible output:

3.141592653589793

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We can use π to calculate the area and circumference of a circle.

12. Circle Area Using math.pi

Formula:

Area = π × r × r

Python program:

```
import math


radius = 5


area = math.pi * radius * radius


print("Area of circle:", area)
```

Output will be approximately:

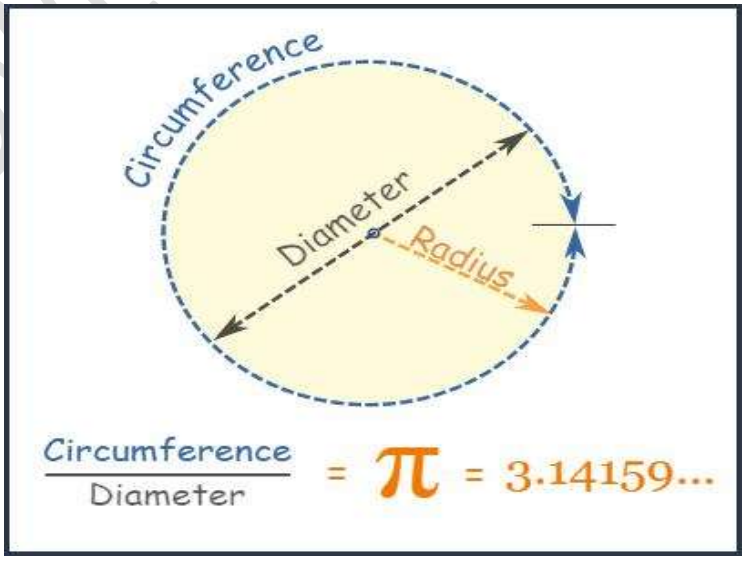
Area of circle: 78.53981633974483

For students, we can display a rounded answer:

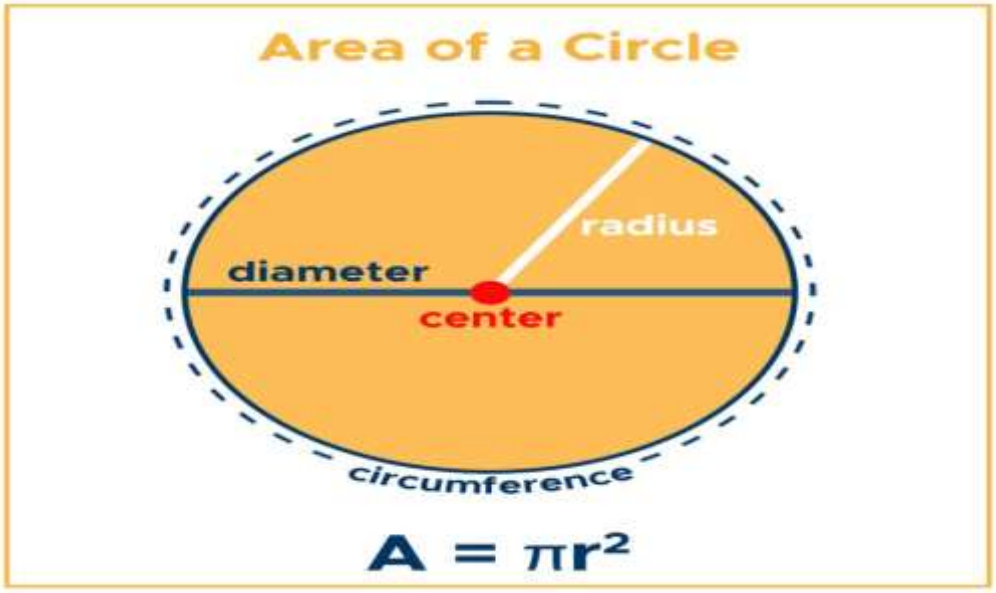
```
print(round(area, 2))
```

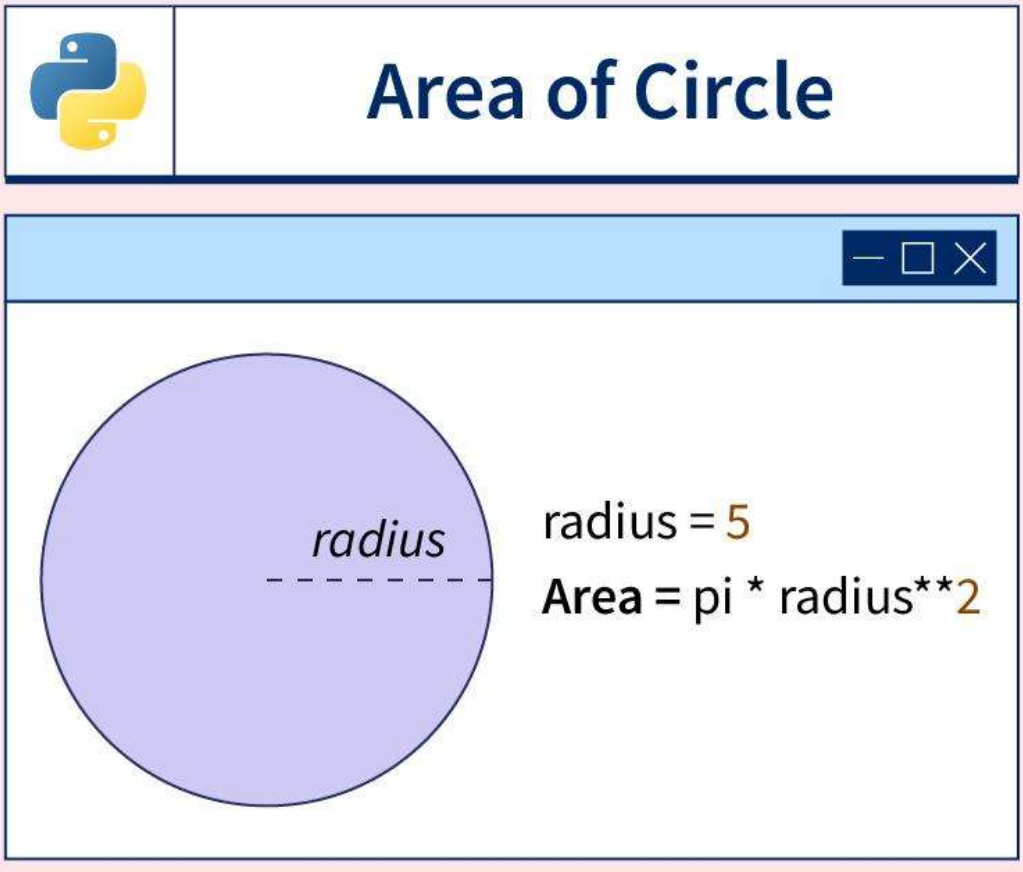
Output:

78.54



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Area of a Circle
radius
diameter
center
circumference
A = πr²


Area of Circle
radius
radius = 5
Area = pi * radius**2

13. Circle Circumference

Formula:

Circumference = 2 × π × r

Python:

```
import math


radius = 5


circumference = 2 * math.pi * radius


print("Circumference:", circumference)
```

14. math.fabs()

fabs() returns the absolute value of a number as a floating-point number.

```
import math


print(math.fabs(-10))
```

Output:

10.0

For beginners, remember:

fabs() removes the negative sign.

15. math.trunc()

trunc() removes the decimal part.

```
import math


print(math.trunc(8.75))
```

Output:

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8

Another example:

print(math.trunc(12.99))

Output:

12

Important Math Functions

Function	Easy Meaning	Example
sqrt()	Square root	sqrt(25)
pow()	Power	pow(2,3)
ceil()	Round up	ceil(4.2)
floor()	Round down	floor(4.8)
factorial()	Factorial	factorial(5)
gcd()	Greatest common divisor	gcd(12,18)
lcm()	Least common multiple	lcm(4,6)
pi	Value of π	math.pi
fabs()	Absolute value	fabs(-5)
trunc()	Remove decimal part	trunc(5.9)

Memory Trick

√ sqrt()

Square Root

⚡ **pow()**

Power

⬆️ **ceil()**

Go Up

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⬇️ **floor()**

Go Down

❗ **factorial()**

Multiply down to 1

🪙 **pi**

Circle calculations

🔗 **gcd()**

Greatest Common Divisor

🔗 **lcm()**

Least Common Multiple

NDATION

Python's Math Module Cheat Sheet

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Number-theoretic functions
1. comb(n, k) — Ways to choose k items from n
2. factorial(n) — n factorial
3. gcd(*integers) — Greatest common divisor

Floating point arithmetic
1. ceil(x) — Round up
2. floor(x) — Round down
3. fabs(x) — Absolute value

Floating point manipulation
1. isclose(a, b) — Check if two values are close
2. isinf(x) — Check if x is infinite
3. isnan(x) — Check if x is NaN

Power, exponential, and logarithmic
1. exp(x) — e^x
2. log(x) — Logarithm (base e)
3. sqrt(x) — Square root of x

Summation and product
1. fsum(iterable) — Precise sum
2. prod(iterable) — Product of elements

Angles
1. degrees(x) — Radians → degrees
2. radians(x) — Degrees → radians

Trigonometry
1. sin(x), cos(x), tan(x) — Sine, cosine, tangent
2. atan2(y, x) — Arc tangent of y/x

Hyperbolic
1. sinh(x), cosh(x), tanh(x) — Hyperbolic functions

Special
1. erf(x) — Error function
2. gamma(x) — Gamma function

Constants
1. pi — π
2. e — Euler's number
3. tau — 2π







Practical Session

Practical 1: Square Root Calculator

```
import math

number = float(input("Enter a number: "))

answer = math.sqrt(number)

print("Square Root:", answer)
```

Example

Enter a number: 64

Square Root: 8.0

Practical 2: Power Calculator

```
import math

number = float(input("Enter number: "))
power = int(input("Enter power: "))

answer = math.pow(number, power)

print("Answer:", answer)
```

Example:

Enter number: 3

Enter power: 2

Answer: 9.0

Practical 3: Ceiling and Floor

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```
import math

number = float(input("Enter a decimal number: "))

print("Ceiling:", math.ceil(number))
print("Floor:", math.floor(number))
```

Example:

Enter a decimal number: 7.4

Ceiling: 8

Floor: 7

Practical 4: Factorial Calculator

```
import math

number = int(input("Enter a number: "))

answer = math.factorial(number)

print("Factorial:", answer)
```

Example:

Enter a number: 5

Factorial: 120

Practical 5: GCD Calculator

```
import math
```

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```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

answer = math.gcd(a, b)

print("GCD:", answer)
```

Example:

Enter first number: 24

Enter second number: 36

GCD: 12

Practical 6: LCM Calculator

```
import math

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

answer = math.lcm(a, b)

print("LCM:", answer)
```

Example:

Enter first number: 4

Enter second number: 6

LCM: 12

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Practical 7: Circle Area Calculator

```
import math

radius = float(input("Enter radius: "))

area = math.pi * radius * radius

print("Area:", area)
```

Practical 8: Circle Circumference

```
import math

radius = float(input("Enter radius: "))

circumference = 2 * math.pi * radius

print("Circumference:", circumference)
```

Practical 9: Absolute Value

```
import math

number = float(input("Enter a number: "))

print("Absolute value:", math.fabs(number))
```

Example:

```
Enter a number: -25

Absolute value: 25.0
```

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Practical 10: Remove Decimal

```
import math

number = float(input("Enter decimal number: "))

print("Number without decimal:", math.trunc(number))
```

Example:

Enter decimal number: 15.89

Number without decimal: 15

Mini Project 1: Math Helper

Let's create a small program that performs several mathematical operations.

```
import math

number = float(input("Enter a number: "))

print("Square Root:", math.sqrt(number))
print("Square:", math.pow(number, 2))
print("Ceiling:", math.ceil(number))
print("Floor:", math.floor(number))
```

Example

Enter a number: 6.4

Square Root: 2.5298221281347035

Square: 40.96

Ceiling: 7

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Floor: 6

Mini Project 2: Circle Calculator

```
import math

radius = float(input("Enter radius: "))

area = math.pi * radius * radius
circumference = 2 * math.pi * radius

print("Area:", round(area, 2))
print("Circumference:", round(circumference, 2))
```

Example

Enter radius: 7

Area: 153.94

Circumference: 43.98

Classroom Activity: Math Challenge

Ask students to predict the answers **before running the program**.

```
import math

print(math.sqrt(36))
print(math.pow(2, 4))
print(math.ceil(5.2))
print(math.floor(9.8))
print(math.factorial(4))
```

Challenge

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Students should write their predicted answers in their notebooks and then run the program to check.

Real-Life Application

The Math Module can be useful in:

- School mathematics
- Construction
- Engineering
- Game development
- Computer programming
- Data calculations
- Science experiments

Common Mistakes

Mistake 1: Forgetting import

Wrong:

```
print(math.sqrt(25))
```

Correct:

```
import math


print(math.sqrt(25))
```

Mistake 2: Wrong spelling

Wrong:

```
math.squrt(25)
```

Correct:

```
math.sqrt(25)
```

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Mistake 3: Using text instead of a number

Wrong:

```
number = input("Enter number: ")


print(math.sqrt(number))
```

input() gives text.

Correct:

```
number = float(input("Enter number: "))


print(math.sqrt(number))
```

Student Practice Questions

Very Short Questions

1. What is a module?
2. What is the Math Module?
3. How do you import the Math Module?
4. What is the use of sqrt()?
5. What does pow() do?
6. What is the use of ceil()?
7. What is the use of floor()?
8. What does factorial() calculate?
9. What is math.pi?
10. What does gcd() mean?

Coding Practice

Question 1

Find the square root of 144.

Question 2

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Calculate 5^2 using math.pow().

Question 3

Find the ceiling and floor of 8.7.

Question 4

Find the factorial of 6.

Question 5

Find the GCD of 20 and 30.

Question 6

Find the LCM of 5 and 10.

Question 7

Calculate the area of a circle with radius 10.

Question 8

Calculate the circumference of a circle with radius 10.

Homework

Create a "Student Math Calculator"

Your program should:

1. Ask the student to enter a number.
2. Display its square root.
3. Display its square.
4. Display its ceiling.
5. Display its floor.
6. Ask for another number.
7. Find the GCD of both numbers.

Hint

```
import math


num1 = float(input("Enter first number: "))
```

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```
print("Square Root:", math.sqrt(num1))
print("Square:", math.pow(num1, 2))
print("Ceiling:", math.ceil(num1))
print("Floor:", math.floor(num1))

num2 = int(input("Enter another number: "))

print("GCD:", math.gcd(int(num1), num2))
```

Lesson Summary

Today we learned:

- 📦 **Module** → A collection of ready-made Python tools.
- 🧮 **Math Module** → Provides mathematical functions.
- import math → Imports the Math Module.
- math.sqrt() → Finds square root.
- math.pow() → Calculates powers.
- math.ceil() → Rounds upward.
- math.floor() → Rounds downward.
- math.factorial() → Calculates factorial.
- math.gcd() → Finds Greatest Common Divisor.
- math.lcm() → Finds Least Common Multiple.
- math.pi → Provides the value of π.
- math.fabs() → Finds absolute value.
- math.trunc() → Removes the decimal part.

Fun Challenge

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Create a program that asks the student for the **radius of a circle** and displays:

- Radius
- Diameter
- Area
- Circumference

Use:

math.pi

for the circle calculations.

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